Final Exam

DDD 2: Introduction to Programming and Basic Data Analysis

Ramesh Beniwal | Admission ID: BAEC2240006

2026-11-03

Table of contents

Setup 1
Load Required Libraries 1
Exam Environment 2
Data Loading 2
QUESTION 1 2
QUESTION 2 5
INTERPRETATION 13

Setup

```
rm(list = ls())
```

Load Required Libraries

```
library(tibble)
library(dplyr)
library(rlang)
library(lobstr)
library(rio)
library(ggplot2)
library(labelled)
library(stargazer)
library(scales)
library(ivo.table)
library(flextable)
library(janitor)
```

```
library(tinytex)
library(labelled)
```

Exam Environment

```
FINAL_EXAM <- new.env()
```

Data Loading

```
library(tidyverse)
library(rio)
#1 HH_DATA
HH_DATA <- import("../data/ddd2/RAW_DATA/ICPSR_36151/HH_DATA/HH_DATA.dta")
View(HH_DATA)
```

QUESTION 1

```
## Creating my own per capita income from household  income variable
# creating household size from given variables

FINAL_EXAM$HH_DATA_q1 <- HH_DATA |>
  mutate(
     family_size = NADULTM + NADULTF + NCHILDM + NCHILDF + NTEENM + NTEENF +
NELDERM + NELDERF,
    income_pc_constructed = INCOME / family_size) |>
  select(INCOME, family_size, income_pc_constructed, INCOMEPC)

# Check for missing values
summary(FINAL_EXAM$HH_DATA_q1)
```

```
     INCOME          family_size     income_pc_constructed    INCOMEPC
 Min.   :-1037040   Min.   : 1.000   Min.   :-289008       Min.   :-289008
 1st Qu.:   39000   1st Qu.: 4.000   1st Qu.:   8000       1st Qu.:   8975
 Median :   73500   Median : 5.000   Median :  15000       Median :  16583
 Mean   :  127760   Mean   : 5.373   Mean   :  26521       Mean   :  29402
 3rd Qu.:  144000   3rd Qu.: 7.000   3rd Qu.:  29421       3rd Qu.:  32400
 Max.   :11360000   Max.   :34.000   Max.   :4161000       Max.   :4161000
```

```r
# Some Oultiers in Income and Per Capita so we need to remove them
#Remove negative values and zeros
FINAL_EXAM$HH_DATA_q1 <- FINAL_EXAM$HH_DATA_q1 |>
  filter(income_pc_constructed > 0, INCOMEPC > 0)


# CHECKING SUMMARY AGAIN TO MATCH
summary(FINAL_EXAM$HH_DATA_q1)
```

```
     INCOME          family_size     income_pc_constructed    INCOMEPC
 Min.   :     100   Min.   : 1.000   Min.   :     16.7     Min.   :     16.7
 1st Qu.:   40230   1st Qu.: 4.000   1st Qu.:   8255.5     1st Qu.:   9193.2
 Median :   75000   Median : 5.000   Median :  15233.3     Median :  16833.3
 Mean   :  129987   Mean   : 5.378   Mean   :  26990.7     Mean   :  29922.6
 3rd Qu.:  145708   3rd Qu.: 7.000   3rd Qu.:  29889.8     3rd Qu.:  32837.1
 Max.   :11360000   Max.   :34.000   Max.   :4161000.0     Max.   :4161000.0
```

```r
# THERE IS DIFFERENCE BETWEEN INCOMEPC AND income_pc_constructed lets know this
difference
FINAL_EXAM$HH_DATA_q1 <- FINAL_EXAM$HH_DATA_q1 |>
  mutate(
    difference = income_pc_constructed - INCOMEPC,
    pct_diff = (difference / INCOMEPC) * 100)

summary(FINAL_EXAM$HH_DATA_q1$difference)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
-825000   -2309       0   -2932       0    4500
```

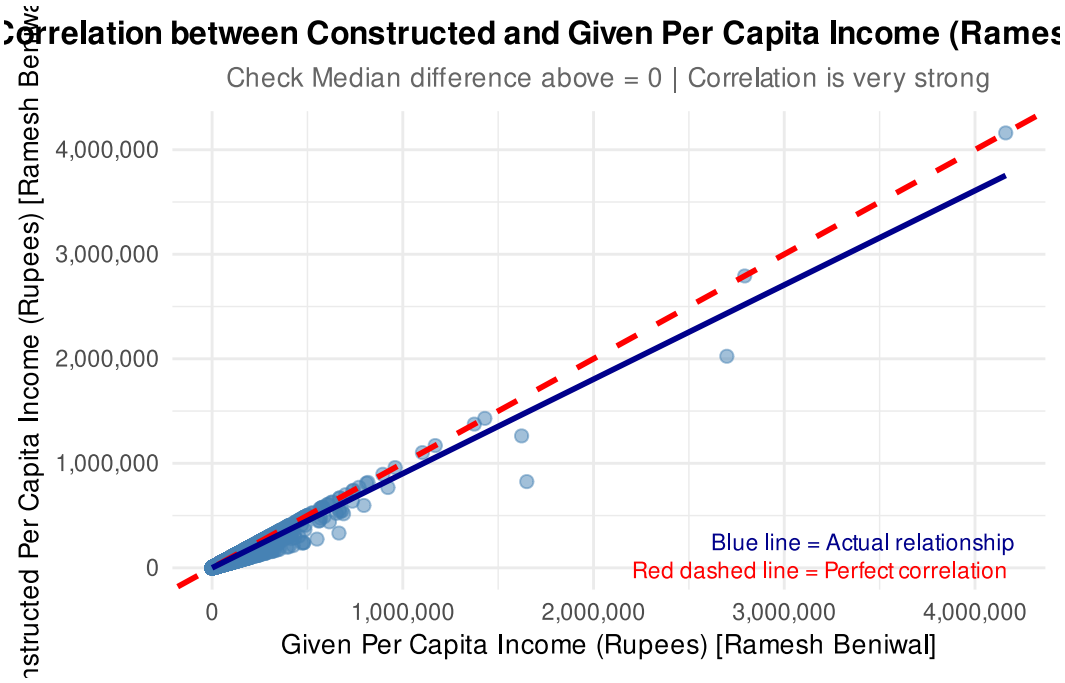
```r
summary(FINAL_EXAM$HH_DATA_q1$pct_diff)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
-50.000 -16.667   0.000  -9.094   0.000  25.000
```

```r
# So if we look into summary of it the median is around 0  for both cases
means our correlation is strong and this difference is may be due to some small
differences in household sizes and rounding of eorrors.


## CORRELATION BETWEEEN THE CONSTRUCTED ADN ORIGINAL INCOMEPC
```

```r
# PLOT CORRELATION BETWEEN CONSTRUCTED AND GIVEN PER CAPITA INCOME
FINAL_EXAM$HH_DATA_q1 |>
  ggplot(aes(x = INCOMEPC, y = income_pc_constructed)) +
  geom_point(alpha = 0.5, size = 2, color = "steelblue") +
   geom_abline(intercept = 0, slope = 1, color = "red", linetype = "dashed",
linewidth = 1) +
  geom_smooth(se = TRUE, method = "lm", color = "darkblue", linewidth = 1) +
  scale_y_continuous(labels = scales::comma) +
  scale_x_continuous(labels = scales::comma) +
   annotate("text", x = Inf, y = -Inf, label = "Red dashed line = Perfect
correlation",
           hjust = 1.1, vjust = -0.5, color = "red", size = 3) +
  annotate("text", x = Inf, y = -Inf, label = "Blue line = Actual relationship",
           hjust = 1.1, vjust = -2, color = "darkblue", size = 3) +
  labs(
    title = "Correlation between Constructed and Given Per Capita Income (Ramesh
Beniwal)",
    x = "Given Per Capita Income (Rupees) [Ramesh Beniwal]",
    y = "Constructed Per Capita Income (Rupees) [Ramesh Beniwal]",
   subtitle = "Check Median difference above = 0 | Correlation is very strong") +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold", size = 12),
    plot.subtitle = element_text(hjust = 0.5, size = 10, color = "gray40"),
    axis.title = element_text(size = 10))
```




```
# coefficient for correlation
FINAL_EXAM$correlation_q1 <- cor(FINAL_EXAM$HH_DATA_q1$INCOMEPC,FINAL_EXAM$HH_DATA_q1$income_pc_co

print(paste("Correlation coefficient (Q1):", round(FINAL_EXAM$correlation_q1,
2)))
```

```
[1] "Correlation coefficient (Q1): 0.98"
```

```
print(paste("Interpretation: Very strong positive correlation - constructed and
given per capita income are highly similar"))
```

```
[1] "Interpretation: Very strong positive correlation - constructed and given
per capita income are highly similar"
```

QUESTION 2

```
# ASSUMPTION - Y GOES UP ~ C GOES UP
# VALIDATION BY RELATIONSHIP BETWEEN COPC AND income_pc_constructed

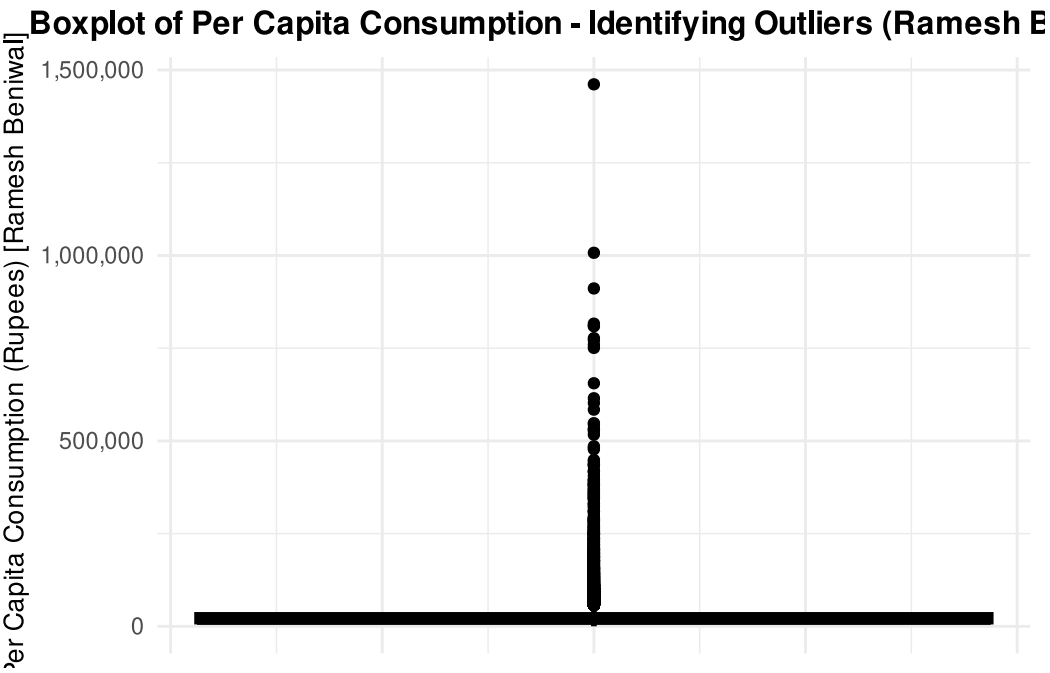
# PART1
FINAL_EXAM$HH_DATA_q2 <- HH_DATA |>
```

```
  mutate(
     family_size = NADULTM + NADULTF + NCHILDM + NCHILDF + NTEENM + NTEENF +
NELDERM + NELDERF,
    income_pc_constructed = INCOME / family_size
  ) |>
  select(income_pc_constructed, INCOMEPC, COPC) |>
  filter(income_pc_constructed > 0, INCOMEPC > 0, COPC > 0)

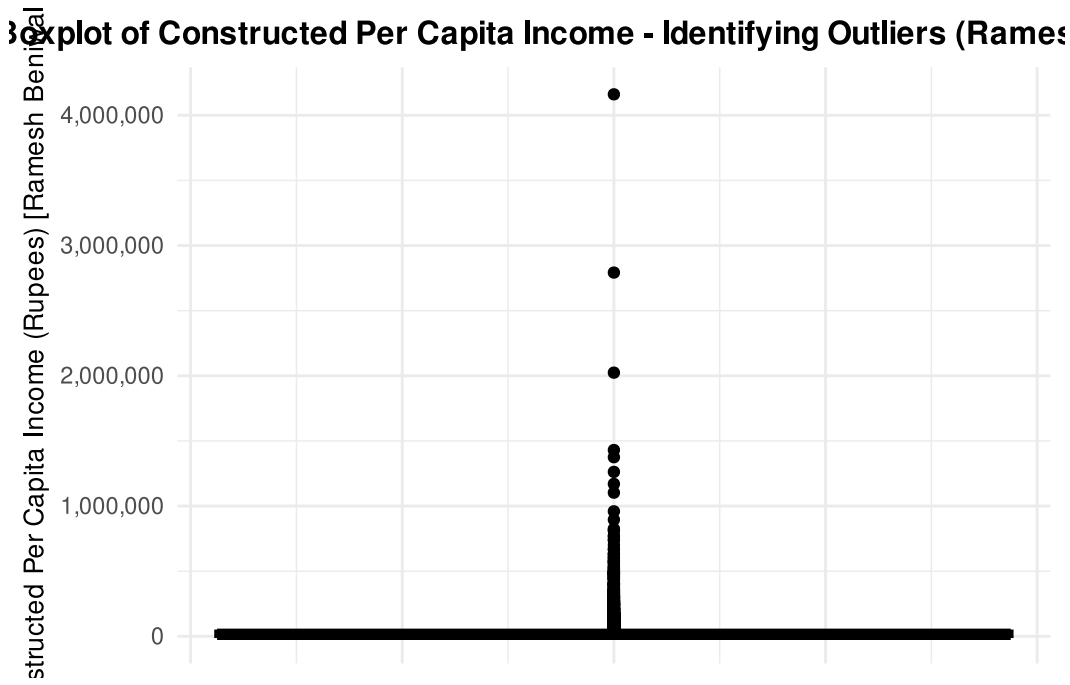
summary(FINAL_EXAM$HH_DATA_q2)
```

```
 income_pc_constructed    INCOMEPC             COPC
 Min.   :     16.7     Min.   :     16.7   Min.   :    180
 1st Qu.:   8258.2     1st Qu.:   9194.2   1st Qu.:  12791
 Median :  15233.7     Median :  16833.3   Median :  19435
 Mean   :  26983.1     Mean   :  29915.6   Mean   :  27071
 3rd Qu.:  29889.8     3rd Qu.:  32849.3   3rd Qu.:  30929
 Max.   :4161000.0     Max.   :4161000.0   Max.   :1461484
```

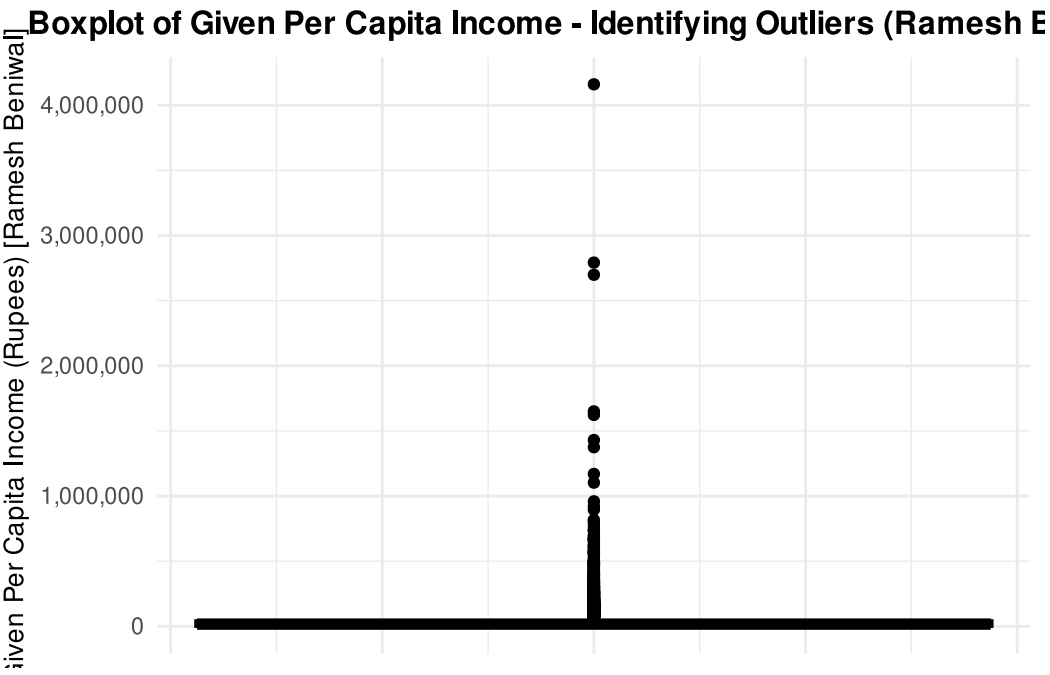
```
#BOXPLOT - Check outliers in COPC before filtering
FINAL_EXAM$HH_DATA_q2 |>
  ggplot(aes(y = COPC)) +
  geom_boxplot(fill = "orange", color = "black", linewidth = 1) +
  scale_y_continuous(labels = scales::comma) +
  labs(
     title = "Boxplot of Per Capita Consumption - Identifying Outliers (Ramesh
Beniwal)",
    y = "Per Capita Consumption (Rupees) [Ramesh Beniwal]") +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold", size = 12),
    axis.title = element_text(size = 10),
    axis.text.x = element_blank())
```




```
# BOXPLOT FOR CONSTRUCTED PER CAPITA INCOME
FINAL_EXAM$HH_DATA_q2 |>
  ggplot(aes(y = income_pc_constructed)) +
  geom_boxplot(fill = "skyblue", color = "black", linewidth = 1) +
  scale_y_continuous(labels = scales::comma) +
  labs(
     title = "Boxplot of Constructed Per Capita Income - Identifying Outliers
(Ramesh Beniwal)",
    y = "Constructed Per Capita Income (Rupees) [Ramesh Beniwal]") +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold", size = 12),
    axis.title = element_text(size = 10),
    axis.text.x = element_blank())
```




```
# BOXPLOT FOR GIVEN PER CAPITA INCOME
FINAL_EXAM$HH_DATA_q2 |>
  ggplot(aes(y = INCOMEPC)) +
  geom_boxplot(fill = "lightgreen", color = "black", linewidth = 1) +
  scale_y_continuous(labels = scales::comma) +
  labs(
    title = "Boxplot of Given Per Capita Income - Identifying Outliers (Ramesh
Beniwal)",
    y = "Given Per Capita Income (Rupees) [Ramesh Beniwal]") +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold", size = 12),
    axis.title = element_text(size = 10),
    axis.text.x = element_blank())
```



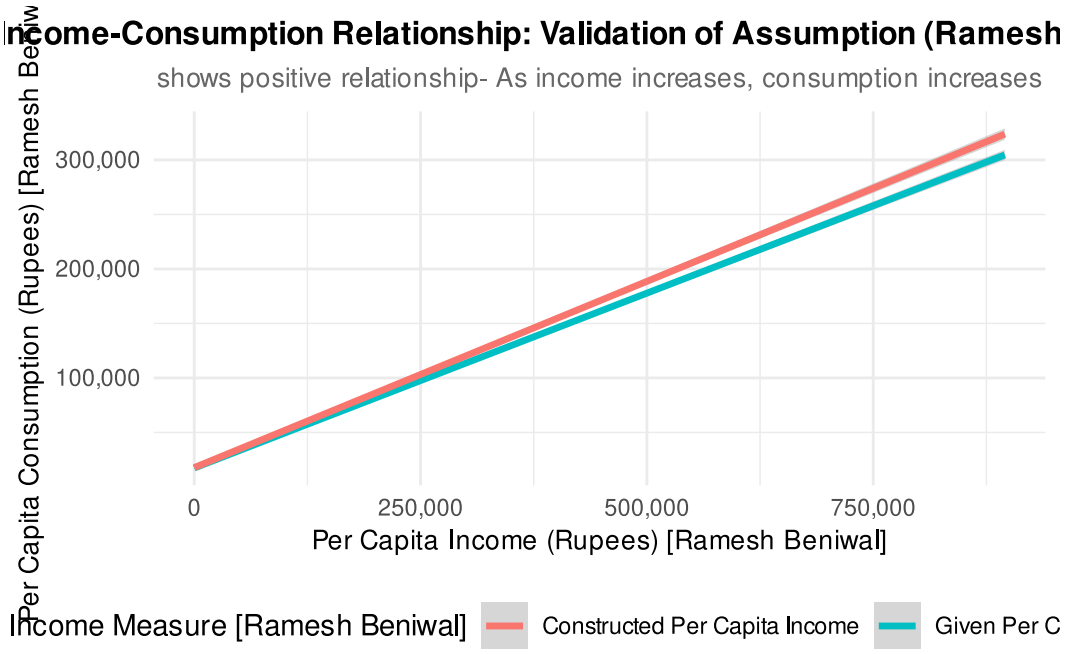

```
# Remove outliers from COPC and other variables
FINAL_EXAM$HH_DATA_q2 <- FINAL_EXAM$HH_DATA_q2 |>
  filter(COPC < 600000, income_pc_constructed < 900000, INCOMEPC < 900000)

summary(FINAL_EXAM$HH_DATA_q2)
```

```
 income_pc_constructed    INCOMEPC             COPC
 Min.   :    16.67     Min.   :    16.67   Min.   :   180
 1st Qu.:  8254.53     1st Qu.:  9192.07   1st Qu.: 12790
 Median : 15225.00     Median : 16825.00   Median : 19426
 Mean   : 26530.25     Mean   : 29398.94   Mean   : 26790
 3rd Qu.: 29841.50     3rd Qu.: 32800.00   3rd Qu.: 30906
 Max.   :895400.00     Max.   :895400.00   Max.   :584348
```

```
# Both relationships on same graph
FINAL_EXAM$HH_DATA_q2 |>
  ggplot() +
  geom_smooth(aes(x = INCOMEPC, y = COPC, color = "Given Per Capita Income"),
              se = TRUE, method = "lm", linewidth = 1.2) +
  geom_smooth(aes(x = income_pc_constructed, y = COPC, color = "Constructed Per
Capita Income"),
              se = TRUE, method = "lm", linewidth = 1.2) +
  scale_x_continuous(labels = scales::comma) +
```

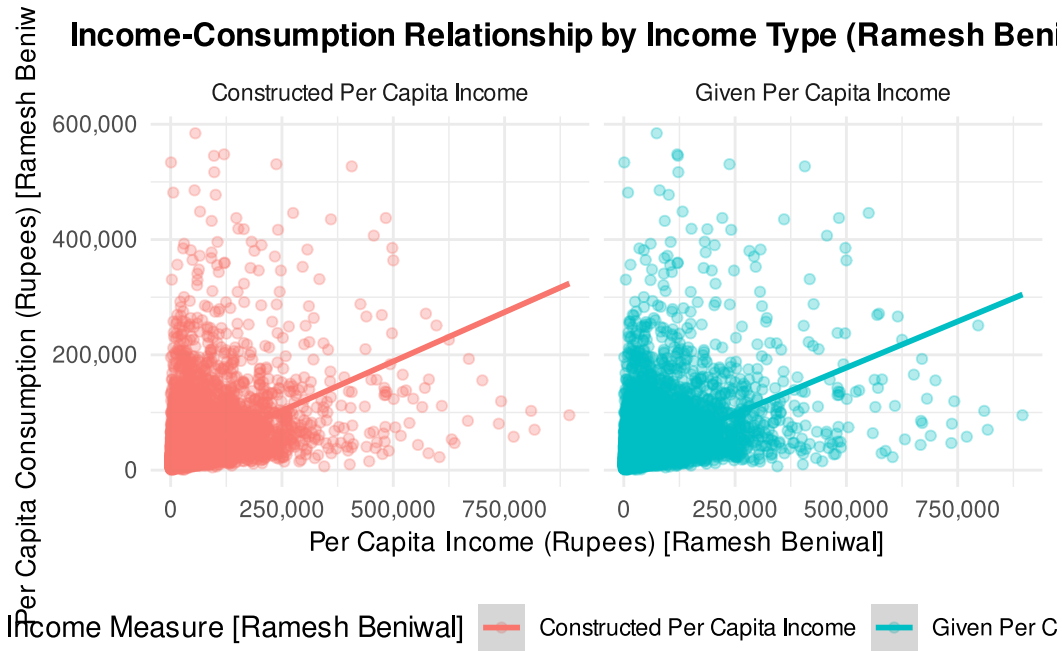
```
  scale_y_continuous(labels = scales::comma) +
  labs(
    title = "Income-Consumption Relationship: Validation of Assumption (Ramesh
Beniwal)",
    x = "Per Capita Income (Rupees) [Ramesh Beniwal]",
    y = "Per Capita Consumption (Rupees) [Ramesh Beniwal]",
    color = "Income Measure [Ramesh Beniwal]",
    subtitle = "shows positive relationship- As income increases, consumption
increases") +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold", size = 12),
    plot.subtitle = element_text(hjust = 0.5, size = 10, color = "gray40"),
    axis.title = element_text(size = 10),
    legend.position = "bottom")
```




```
# side by side
FINAL_EXAM$HH_DATA_q2_side<- FINAL_EXAM$HH_DATA_q2 |>
  pivot_longer(cols = c(INCOMEPC, income_pc_constructed),
               names_to = "income_type",
               values_to = "income_value") |>
  mutate(income_type = case_when(
    income_type == "INCOMEPC" ~ "Given Per Capita Income",
```

```
    income_type == "income_pc_constructed" ~ "Constructed Per Capita Income"))

FINAL_EXAM$HH_DATA_q2_side|>
  ggplot(aes(x = income_value, y = COPC, color = income_type)) +
  geom_point(alpha = 0.3, size = 1.5) +
  geom_smooth(se = TRUE, method = "lm", linewidth = 1) +
  facet_wrap(~income_type, scales = "free_x") +
  scale_x_continuous(labels = scales::comma) +
  scale_y_continuous(labels = scales::comma) +
  labs(
    title = "Income-Consumption Relationship by Income Type (Ramesh Beniwal)",
    x = "Per Capita Income (Rupees) [Ramesh Beniwal]",
    y = "Per Capita Consumption (Rupees) [Ramesh Beniwal]",
    color = "Income Measure [Ramesh Beniwal]") +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold", size = 12),
    axis.title = element_text(size = 10),
    legend.position = "bottom")
```




```
#Cor calculation for both
FINAL_EXAM$cor_given_income       <-       cor(FINAL_EXAM$HH_DATA_q2$INCOMEPC,
FINAL_EXAM$HH_DATA_q2$COPC)
```

```
FINAL_EXAM$cor_constructed_income                                    <-
cor(FINAL_EXAM$HH_DATA_q2$income_pc_constructed, FINAL_EXAM$HH_DATA_q2$COPC)
```

```
# Results / Print
print(paste("Correlation (Given Per Capita Income vs Consumption):",
round(FINAL_EXAM$cor_given_income, 2)))
```

```
[1] "Correlation (Given Per Capita Income vs Consumption): 0.49"
```

```
print(paste("Correlation (Constructed Per Capita Income vs Consumption):",
round(FINAL_EXAM$cor_constructed_income, 2)))
```

```
[1] "Correlation (Constructed Per Capita Income vs Consumption): 0.48"
```

```
print("INTERPRETATION: ASSUMPTION VALIDATED - Both income measures show POSITIVE
correlation with consumption")
```

```
[1] "INTERPRETATION: ASSUMPTION VALIDATED - Both income measures show POSITIVE
correlation with consumption"
```

```
print("As per capita income increases, per capita consumption also increases")
```

```
[1] "As per capita income increases, per capita consumption also increases"
```

INTERPRETATION

ℹ Key Findings

Correlation Coefficients: - **Given Per Capita Income vs Consumption:** 0.49 - **Constructed Per Capita Income vs Consumption:** 0.48

Interpretation:

Both per capita income measures (manually and given) show a **POSITIVE correlation** (approximately 0.48-0.49) with per capita consumption, which **VALIDATES the assumption** that as income increases, consumption also increases.

Key Observations:

1. **Positive Relationship:** The correlation of ~0.48 indicates a positive relationship between income and consumption.
2. **Both Measures Consistent:** The constructed per capita income and given per capita income show nearly identical correlations (0.49 vs 0.48), confirming our constructed income variable is valid.
3. **Assumption Validated:** The economic assumption that consumption increases with income is **CONFIRMED** as per Household Data we are using.

Conclusion:

The analysis provides evidence supporting the economic thinking and assumption of positive income-consumption relationship. Both graphs and correlation shows that their is positive relationships between them and positively related to each other.